

## BIVARIATE ANALYSIS

### 1. COVARIANCE:

|          | sl_no         | ssc_p        | hsc_p        | degree_p     | etest_p       | mba_p        | salary        |
|----------|---------------|--------------|--------------|--------------|---------------|--------------|---------------|
| sl_no    | 3870.000000   | -52.641355   | -58.106028   | -40.413645   | 52.556168     | 8.102336     | 2.484009e+05  |
| ssc_p    | -52.641355    | 117.228377   | 60.348373    | 42.897137    | 37.659225     | 24.535952    | 1.976764e+04  |
| hsc_p    | -58.106028    | 60.348373    | 118.755706   | 34.819820    | 35.461678     | 22.555846    | 4.600806e+04  |
| degree_p | -40.413645    | 42.897137    | 34.819820    | 54.151103    | 21.929469     | 17.272020    | -8.064357e+03 |
| etest_p  | 52.556168     | 37.659225    | 35.461678    | 21.929469    | 176.251018    | 16.886973    | 1.571579e+05  |
| mba_p    | 8.102336      | 24.535952    | 22.555846    | 17.272020    | 16.886973     | 34.028376    | 6.611551e+04  |
| salary   | 248400.921950 | 19767.643976 | 46008.057275 | -8064.357161 | 157157.850783 | 66115.509283 | 5.999726e+09  |

**Question:** What kind of covariance between degree\_p and etest\_p?

**Ans:** Positive covariance between degree\_p and etest\_p. This value is 21%.

**Question:** What kind of covariance between etest\_p and mba\_p?

**Ans:** Positive covariance between etest\_p and mba\_p. This value is 16%.

### 2. CORRELATION

|          | sl_no     | ssc_p     | hsc_p     | degree_p  | etest_p  | mba_p    | salary    |
|----------|-----------|-----------|-----------|-----------|----------|----------|-----------|
| sl_no    | 1.000000  | -0.078155 | -0.085711 | -0.088281 | 0.063636 | 0.022327 | 0.051550  |
| ssc_p    | -0.078155 | 1.000000  | 0.511472  | 0.538404  | 0.261993 | 0.388478 | 0.023571  |
| hsc_p    | -0.085711 | 0.511472  | 1.000000  | 0.434206  | 0.245113 | 0.354823 | 0.054506  |
| degree_p | -0.088281 | 0.538404  | 0.434206  | 1.000000  | 0.224470 | 0.402364 | -0.014148 |
| etest_p  | 0.063636  | 0.261993  | 0.245113  | 0.224470  | 1.000000 | 0.218055 | 0.152829  |
| mba_p    | 0.022327  | 0.388478  | 0.354823  | 0.402364  | 0.218055 | 1.000000 | 0.146324  |
| salary   | 0.051550  | 0.023571  | 0.054506  | -0.014148 | 0.152829 | 0.146324 | 1.000000  |

**Question:** What kind of correlation between degree\_p and etest\_p?

**Ans:** Positive correlation between degree\_p and etest\_p. This value is 0.22%.

**Question:** What kind of correlation between etest\_p and mba\_p?

**Ans:** Positive correlation between etest\_p and mba\_p. This value is 0.21%.

### 3. MULTICOLLINEARITY:

**Ques: How can we handle Multicollinearity?**

**Answer:**

When multicollinearity is detected in regression model, there are several ways to address it. Some common solutions:

- **Remove Highly Correlated Predictors:**
  - If two or more variables are highly correlated, consider removing one of them.
  - This helps reduce redundancy and simplifies the model without losing too much information.
- **Combine Variables (Feature Engineering):**
  - **Principal Component Analysis (PCA)**: PCA reduces the dimensionality of the dataset by combining correlated predictors into a smaller number of uncorrelated components.
  - This can help in retaining most of the explanatory power while eliminating multicollinearity.
- **Factor Analysis:**
  - Another dimensionality reduction technique that can combine correlated variables into latent factors.
- **Use Regularization:**
  - **Ridge Regression** (L2 regularization): This method adds a penalty to the size of the coefficients, effectively reducing the impact of correlated predictors by shrinking their values.
  - **Lasso Regression** (L1 regularization): Lasso not only shrinks coefficients but also forces some coefficients to zero, effectively removing less important predictors. This can help in selecting the most relevant variables and reducing multicollinearity.
- **Increase the Sample Size:**
  - Increasing the sample size can reduce the variance of the coefficient estimates and improve the reliability of the regression model.
  - More data can help minimize the effect of multicollinearity by providing more variation among the predictors.
- **Center the Data (Standardization):**
  - In some cases, centering the data (subtracting the mean) can help reduce multicollinearity.
  - It doesn't always eliminate the problem, but it can make the interpretation of the coefficients more meaningful by reducing correlation between the predictors.

- **Stepwise Selection (Forward or Backward):**

- **Stepwise selection** can help you identify the most important features while potentially dropping variables that cause high multicollinearity.
- This is done iteratively by adding or removing features based on their contribution to the model.

#### **4. What is the purpose of Homoscedasticity and heteroscedasticity? When do use this? Which is best?**

- **Homoscedasticity:**

- Homoscedasticity means the **variance of errors remains constant** for all values of the independent variable.
- Homoscedasticity helps for:
  - Regression predictions are trustworthy
  - Statistical tests (t-test, F-test) are valid
  - Confidence intervals are accurate
  - The model performs consistently for all observations
- Use homoscedasticity when:
  - Error variance is expected to stay constant.
  - Applying standard linear regression assumptions.
  - For reliable:
    - hypothesis testing,
    - confidence intervals,
    - p-values.

- **Heteroscedasticity:**

- Heteroscedasticity means the **variance of errors changes** at different levels of the independent variable.
- Detecting heteroscedasticity helps because:
  - It shows the regression assumptions are violated.
  - Statistical tests may become misleading.
  - Standard errors can become incorrect.
  - Predictions may be less reliable.
- Use homoscedasticity when:
  - Error variance changes with data size.
  - Large values show larger variability.
  - Regression assumptions may be violated.

## CONCLUSION:

In regression analysis, **homoscedasticity** is generally considered the better.

Because it gives:

- Accurate statistical results
- Reliable p-values and confidence intervals
- Better hypothesis testing
- Stable prediction errors
- Proper linear regression assumptions